

AAKASH SIRICILLA

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EDUCATION

New Jersey Institute of Technology, Master of Science in Computer Science

Sep 2023 – May 2025

Coursework: Machine Learning & MLOps, Deep Learning, Distributed Systems

GPA: 3.93/4.0

Kommuri Pratap Reddy Institute of Technology, B.Tech. Computer Science & Engineering

Aug 2019 – May 2023

Coursework: Cloud Computing, Data Structures & Algorithms, Operating Systems, Database Systems

GPA: 8.55/10

TECHNICAL SKILLS

AI & LLM Systems: LLM Agents (planning, tool use, computer use), RAG, Semantic Search, Embeddings, Prompt Engineering, OpenAI/OpenRouter APIs, MLflow, Vision-Guided Automation

Languages: Python, Java, TypeScript, SQL, C/C++

Backend & APIs: FastAPI, Spring Boot, REST APIs, WebSockets, Celery, Async Processing, JWT Authentication

Databases & Vector Stores: PostgreSQL, MySQL, Redis, Qdrant, FAISS, TimescaleDB, SQLite

Frontend & Apps: React, Next.js, Electron, Chrome Extensions (MV3), Tailwind CSS

Cloud & DevOps: Docker, Kubernetes, AWS, CI/CD, GitHub Actions, Prometheus, Grafana

WORK EXPERIENCE

Spectrum Capital NJ

May 2025 – Present

Software Engineer, Distributed Systems

New Jersey, USA

- Engineered centralized financial-operations dashboards (Java, Spring Boot, React, PostgreSQL) that consolidated portfolio tracking and client workflows, cutting manual reconciliation effort by 40%.
- Optimized high-throughput REST APIs and complex PostgreSQL query pipelines, reducing average response latency by 35% and raising throughput for data-intensive internal services.
- Automated end-to-end financial data ingestion and validation with Python/SQL ETL pipelines, accelerating report generation by 60% while ensuring downstream data integrity.
- Architected role-based workflow orchestration and audit-tracking modules with granular access controls and transactional logging, slashing issue-investigation time by 45%.
- Containerized and deployed Spring Boot backend services on AWS with Docker, standardizing deployment environments and reducing release inconsistencies.

New Jersey Institute of Technology

Mar 2024 – May 2025

Research Assistant

Newark, NJ, USA

- Optimized PostgreSQL and MySQL queries across 120K+ records by redesigning joins and indexing strategies, cutting query execution time by 40%.
- Automated Python-based ETL and preprocessing pipelines for recurring datasets, reducing manual processing effort by 60% and improving reliability of downstream ML and reporting workflows.
- Built data pipelines and dashboards tracking operational and model-performance metrics, shortening reporting turnaround by 30%.

Cognier Insights

Apr 2021 – Sep 2023

Software Engineer

Hyderabad, India

- Developed customer-facing CRM modules and workflow automation (Java, Spring Boot, React, PostgreSQL) streamlining lead management and client onboarding, reducing manual coordination effort by 35%.
- Designed REST APIs, service-layer validation, and PostgreSQL-backed data access for data-intensive workflows, improving retrieval performance by 25% and reliability of transaction flows.
- Created reusable React components integrated with Spring Boot services, cutting feature delivery time by 30%.

PROJECTS

Astra – Autonomous Computer-Use AI Agent | Python, FastAPI, Electron, Next.js, Chrome MV3, SQLite

- Architected an LLM-driven autonomous agent that completes real-world digital tasks end-to-end (job applications, bookings, form filling) through a vision-action loop — screenshot → single-action model decision → execution → read-back verification — with step-level audit receipts and LLM-curated persistent memory.
- Built dual execution runtimes — a Chrome MV3 extension for DOM-verified browser automation and a Windows desktop controller (UIA/SendInput) — with per-task routing and an LLM-classified chat ingress routing requests to tasks, clarifications, or memory.
- Engineered safety-first agent architecture: approval gates before any submit/pay action, a credential vault isolated from the model, sensitive-field masking, and sanctioned-URL navigation — backed by a FastAPI + SQLite backend with ~1,200 automated tests.

DocuMindAI – Enterprise Document Intelligence (RAG) Platform | FastAPI, React, Qdrant, Redis, Celery, MLflow | [GitHub](#)

- Developed a full-stack RAG platform with asynchronous ingestion, hybrid (semantic + keyword) retrieval over Qdrant embeddings, and contextual chat, reducing document lookup time by nearly 70%.
- Implemented background task execution with Celery and Redis and modular service orchestration supporting multi-document workloads, with MLflow-tracked experiments across retrieval configurations.

SentinelAI – Real-Time Network Monitoring Platform | Python, FastAPI, Redis, TimescaleDB, Next.js, Docker | [GitHub](#)

- Engineered a real-time event-processing pipeline with streaming analytics, Redis-backed state, and sub-500ms request handling for high-volume anomaly-detection workflows, plus a WebSocket-powered live monitoring dashboard with replay controls and analytics views.

SyncStream – Real-Time Collaborative Streaming Platform | Java, Spring Boot, Next.js, WebSockets, PostgreSQL | [GitHub](#)

- Built a Dockerized full-stack real-time collaboration platform with invite-based sessions, persistent room state, and low-latency user synchronization via Spring Boot WebSockets.